

Lecture 10 - Regression Discontinuity

Alice Wu

UW Madison Econ 695

- RD Framework
 - Sharp Identification
 - Fuzzy Identification
- Empirical Examples:
 - Medicare (running variable = age)
 - Access to student loan and college enrollment in Chile (running variable = test score)
 - Economics major and earnings from UCSC (running variable = GPA)

- Interested in “causal effect” of x on y , holding other things constant.
- Gold standard: randomized experiment (RCT)
- **Non-experimental methods:**
 - 1 OLS and reweighting counterfactuals (Oaxaca; DFL)
 - 2 IV
 - 3 Diff in diff
 - 4 **Regression Discontinuity**

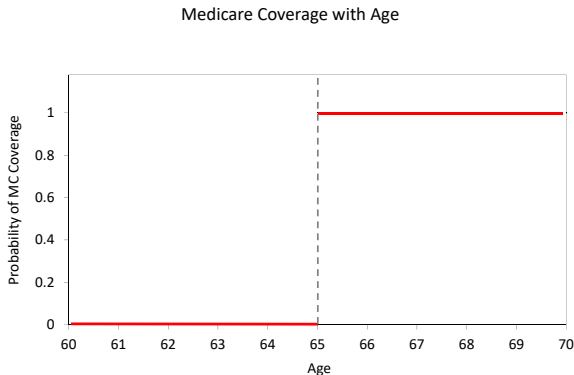
Regression Discontinuity: General Ideas

- RD arises when assignment (D_i) is based on a **discontinuous function** of some “random” or “as good as random” *running variable* x_i . For example, a sharp RD takes the form:

$$D_i = \begin{cases} 0 & x < c \\ 1 & x \geq c \end{cases}$$

- Units on either side of the discontinuity are “the same”.* The distribution of r.v. and other observable characteristics should be smooth around the cutoff.
- Examples:
 - election outcomes (rv=vote share). Let D_i indicate being elected, $D_i = 1[x_i \geq 0.50]$ based on majority rule.
 - admission or financial aid based on test scores (rv=score)
 - rules based on job tenure, duration of marriage etc (rv=duration)
 - rules based on age (rv=age). Let D_i indicate eligibility for Medicare, $D_i = 1[x_i \geq 65]$ for rv $x = \text{age}$.

Regression Discontinuity: General Ideas

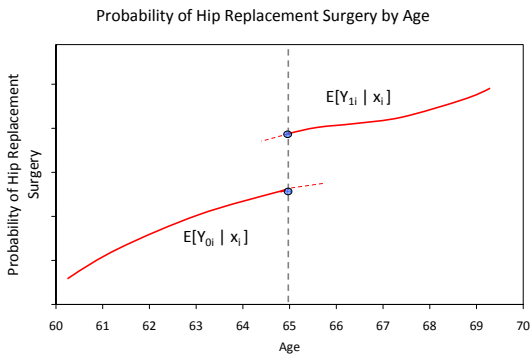


We could exploit this jump in insurance coverage to look at changes in health care usage, and health outcomes...

- x_i = running variable
- D_i = assignment status: $D_i = 1$ if “treated” otherwise 0
- *Sharp RD*: $D_i = 0$ if $x_i < c$; $D_i = 1$ if $x_i \geq c$, 0 otherwise.
 - note: can always normalize x and set $c = 0$
- Potential outcomes $Y_i(0)$ if $D_i = 0$; $Y_i(1)$ if $D_i = 1$
- *Key assumption*:
 - $E[Y_i(0)|x_i]$ is continuous function of x at $x = 0$ (the cutoff)
 - $E[Y_i(1)|x_i]$ is continuous function of x at $x = 0$

RD Framework

- Medicare example: given a cross-sectional data with some people below age $x_i = 65$ and some above 65, you could look at the changes in $Y_i = \text{Hip Replacement}$:



- Define

$$\beta_1 = E[Y_i(1) - Y_i(0) | x_i = 0]$$

This is the treatment effect on agents with $x = 0$ (at the cutoff). Notice that for $x_i < 0$ we have realized $Y_i = Y_i(0)$, and for $x_i \geq 0$ we have $Y_i = Y_i(1)$. So:

- we can estimate $E[Y_i(0)|x_i]$ using data for $x < 0$
- we can estimate $E[Y_i(1)|x_i]$ using data for $x \geq 0$

Then we compare these estimates at $x = 0$ and form $\hat{\beta}_1$.

RD Framework - Example 1

- Consider this special case (linear CEF + constant treatment effect $Y_i(1) - Y_i(0) \equiv \beta_1$):

$$\begin{aligned}E[Y_i(0)|x] &= \alpha_0 + \alpha_1 x \\E[Y_i(1)|x] &= E[Y_i(0)|x] + \beta_1\end{aligned}\tag{1}$$

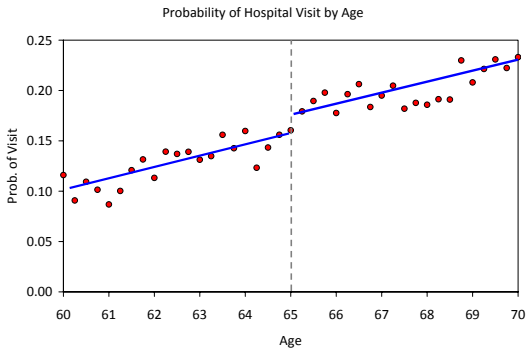
- In this case we can estimate a very simple model:

$$\begin{aligned}Y_i &= \begin{cases} \alpha_0 + \alpha_1 x_i + u_{0i} & \text{if } x_i < 0 \\ \alpha_0 + \alpha_1 x_i + \beta + u_{1i} & \text{if } x_i \geq 0 \end{cases} \\&= \alpha_0 + \alpha_1 x_i + \beta_1 D_i + u_i \text{ for } D_i = 1[x_i \geq 0]\end{aligned}\tag{2}$$

Based on the model (1), we have $E[u_{0i}|x_i] = 0$, $E[u_{1i}|x_i] = 1$, and $u_i = D_i u_{1i} + (1 - D_i) u_{0i}$ is also IND of x_i . What does that imply for our sample OLS?

- Note in this simple model, the slope on x_i is α_1 below and above the cutoff.

RD Framework - Example 1



- Slightly more general model (slope changes at $x = 0$)

$$\begin{aligned} Y_i &= \begin{cases} \alpha_0 + \alpha_1 x_i + u_{0i} & \text{if } x_i < 0 \\ \gamma_0 + \gamma_1 x_i + u_{1i} & \text{if } x_i \geq 0 \end{cases} \\ &= \alpha_0 + \underbrace{(\gamma_0 - \alpha_0)}_{\beta_1} D_i + \alpha_1 x_i + (\gamma_1 - \alpha_1) D_i x_i + u_i \\ &= \beta_0 + \beta_1 D_i + \beta_2 x_i + \beta_3 D_i x_i + u_i \end{aligned} \quad (3)$$

(3) satisfy:

$$\begin{aligned} \beta_2 &= \alpha_1 \text{ slope when } x < 0 \\ \beta_2 + \beta_3 &= \gamma_1 \text{ slope when } x \geq 0 \end{aligned}$$

- Could also have quadratic x^2 (or cubic x^3) with different coefficients on each side of $x = 0$ cutoff. Note critical assumption that $c = 0$ (e.g., define $x = \text{age} - 65$, the cutoff for Medicare is 0).

Identifying Assumptions

- **Identifying assumption:**

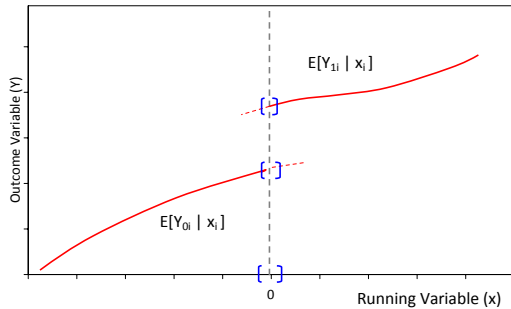
$E[Y_i(0)|x_i]$ and $E[Y_i(1)|x_i]$ are continuous functions of x at “discontinuity” $x = 0$. What does it mean?

- Consider sample 1 with $x_i \in (-\theta, 0)$ and associated $(Y_i(0), Y_i(1))$ vs. sample 2 with $x_i \in [0, \theta)$ and associated $(Y_i(0), Y_i(1))$
 - $\Rightarrow$ distributions of $Y_i(0)$ from 2 samples are centered at $\sim$ same mean
 - and distributions of $Y_i(1)$ from 2 samples are centered at $\sim$ same mean

$\Rightarrow$ *Can use $Y_i(0)$ for sample 1 as “control group” for $Y_i(1)$'s observed in sample 2.*

Identifying Assumptions

Continuity Assumptions of RD Model



Identifying Assumptions

- A sufficient condition for the required continuity is that around $x = 0$, individual assignment to the plus ($x > 0$) or minus ($x < 0$) side is **as good as random**.
- This condition has a **testable implication**:
 - if w_i is some predetermined characteristic (determined prior to assignment), then $E[w_i|x_i]$ will be continuous at $x = 0$.
 - we also expect the relative number of observations to trend smoothly at $x = 0$. No bunching/manipulation around cutoff.

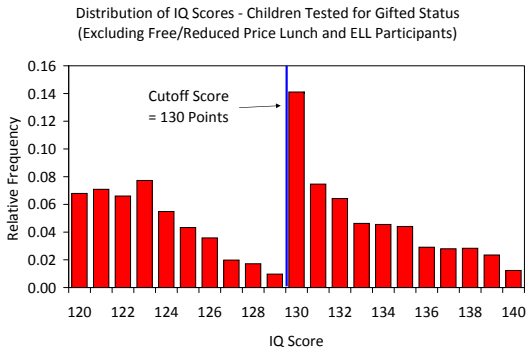
Identifying Assumptions

- How could the RD assumptions break down? Think of the case where $D_i = 1$ is desirable (you win a scholarship if your test score is above a cutoff). If people can bribe the test-taker – or take the test again – we would expect to see:
 - (a) differences in characteristics of people on either side of the cutoff
 - (b) a “spike” of people just to the right

Intuitively, if assignment to the plus or minus side is as good as random, we can think of the people **on either side as treatments and controls in a randomized experiment!**

Identifying Assumptions

- Researchers therefore look for evidence of “manipulation” of the running variable.



Example: Credit Access and College Enrollment

- Alex Solis (2017) asks “Does access to credit explain the gap in schooling attainment between children from richer and poorer families?”. Measuring the effects of credit constraints on college enrollment is challenging:
 - hard to determine whether a family has access to credit;
 - many other **unobservables and self-selection** that affect college enrollment that are likely to be correlated with access to credit -> biased estimates. For example, kids from richer families also have (a) better access to credit, and (b) stronger preferences and preparation for college education.
- He exploits a sharp eligibility rules for college loans in Chile and applies a sharp RD design to estimate the causal impact of **credit access on college enrollment and persistence**.

Example: Credit Access and College Enrollment

- Setting:
 - in Chile, students take “PSU” test (96% of HS grads take the test)
 - colleges and universities use PSU + GPA to rank students and admit
 - students with family taxable income < 75th percentile can use a loan program if they score **at least 475 points** (combined math and language) on PSU.
 - -sample = 475,000 students who wrote the test 2007-2009.
Administrative data followup to measure enrollment in first year of college/university.
- “Crossing the 475-point PSU test threshold implies a sharp change in access to tuition loans.”

running var: $score_i$

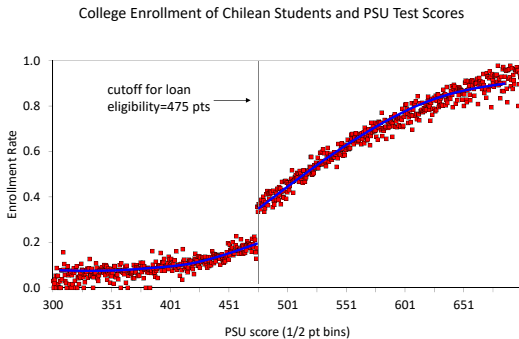
eligibility for loan: $D_i = 1[score_i \geq 475]$

Example: Credit Access and College Enrollment

- Intuition for identification:
 - PSU score has a random component with a continuous density
 - $\Rightarrow$ the probability of scoring ϵ above the cutoff or scoring ϵ below is the same (for a small ϵ). In other words, eligibility for treatment (credit access) around the cutoff will be as good as randomly assigned (Lee 2008).
- Under the identifying assumption, compare students just below vs. just above the cutoff. Learn $E[Y_i(0)|\epsilon \text{ above}]$ from $E[Y_i(0)|\epsilon \text{ below}]$.

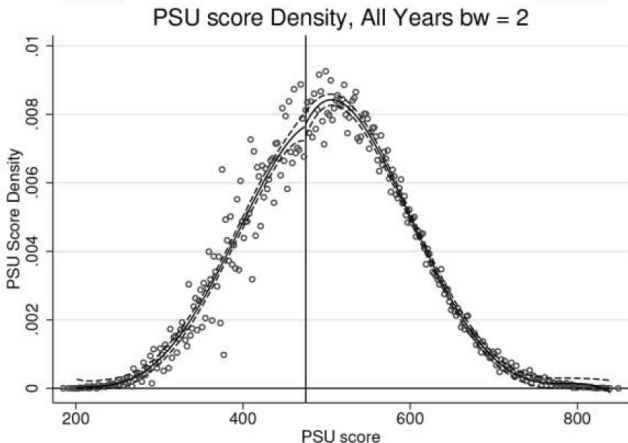
Example: Credit Access and College Enrollment

- Is there a jump in college enrollment?



Example: Credit Access and College Enrollment

- Examine the identifying assumptions for RD:
 - Manipulation of scores around cutoff?



Example: Credit Access and College Enrollment

- Jumps in other characteristics around cutoff?

TABLE
BALANCE OF COVARIATES: POPULATION

VARIABLE	POPULATION				
	Mean (1)	SD (2)	Level (3)	Diff. (4)	SE (5)
Student characteristics:					
High school GPA	56.30	8.54	54.0	-.191	(.100)*
Public school	.45	.50	.44	.005	(.005)
Voucher school	.52	.50	.52	-.005	(.005)
Private school	.03	.17	.04	.000	(.002)
Female	.57	.50	.54	.009	(.005)*
Parent education:					
Mother years of education	11.09	3.51	10.95	-.013	(.038)
Father years of education	11.12	3.73	11.02	-.003	(.042)
Mother dropout high school	.41	.49	.44	-.001	(.005)
Father dropout high school	.44	.50	.45	-.002	(.005)
Mother college graduate	.07	.25	.06	.003	(.003)
Father college graduate	.07	.26	.07	.000	(.003)

Example: Credit Access and College Enrollment

- Concerns about those who take the PSU test multiple times? Those who score below 475 the first time may retake the exam next year and become eligible for student loan in the future
 $\Rightarrow$ eligibility for loan $D_i \neq 1[\text{score}_i \geq 475]$.
- He uses a fuzzy RD:

running var: $x_i = \text{test score on the } \mathbf{1st\ attempt} - 475$

instrument: $z_i = 1[x_i \geq 0]$

$D_i = \mathbf{ever}$ being eligible for loans

The ratio (reduced-form over first-stage):

$$\frac{\lim_{x \rightarrow 0^+} Y_i - \lim_{x \rightarrow 0^-} Y_i}{\lim_{x \rightarrow 0^+} D_i - \lim_{x \rightarrow 0^-} D_i}$$

is a local estimate of the impact of credit access on college enrollment, **local to the margin** - score around 475.

Question: What is the denominator if we have a sharp RD?

Example: Credit Access and College Enrollment

- This works as long as not everyone who score just below 475 retake the exam AND are able to score above 475. $\Rightarrow$ “There will still be a discontinuous jump in the ultimate availability of loans at the threshold”. $\lim_{x \rightarrow 0^+} D_i - \lim_{x \rightarrow 0^-} D_i > 0$.
- We will now look at Fuzzy RD in detail.

- In many cases, we don't have a deterministic assignment rule. Instead the probability of assignment to “treatment” depends on the running variable x and jumps discontinuously at $x = 0$. This leads to *Fuzzy RD*:
- *Assignment/First-stage*:

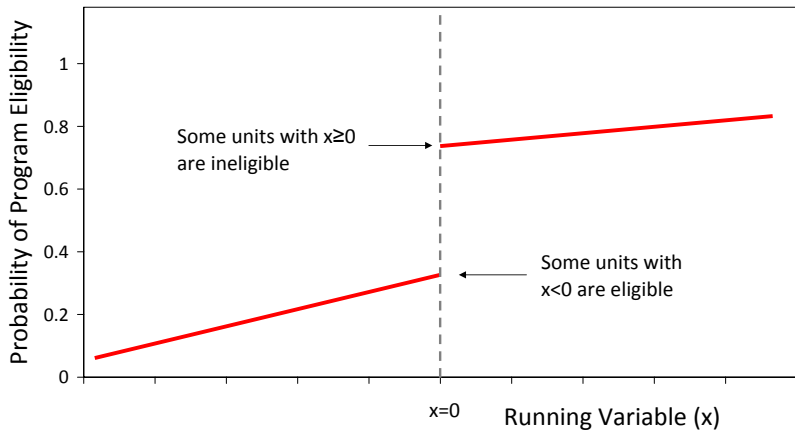
$$Pr(D_i = 1|x_i) = \begin{cases} f_0(x_i) & \text{for } x_i < 0 \\ f_1(x_i) & \text{for } x_i \geq 0 \end{cases}$$

with $f_1(0) \neq f_0(0)$

- Define $z_i = 1[x_i \geq 0]$. (note that $z_i = D_i$ in the 'sharp' case). Then

$$Pr(D_i = 1|x_i) = f_0(x_i) + (f_1(x_i) - f_0(x_i)) z_i$$

Fuzzy RD Based on Discontinuity in Program Eligibility



Identifying Assumptions for Fuzzy RD

- *Key assumptions for fuzzy RD:*
 - ① $E[Y_i(0)|x_i]$ and $E[Y_i(1)|x_i]$ are continuous functions of x at $x = 0$ (threshold).
 - ② First stage: $f_1(0) - f_0(0) \neq 0$, f_1 continuous from above, f_0 continuous from below.
- Define:

$$\begin{aligned} \text{(reduced form)} \quad \delta_1 &= \lim_{x \rightarrow 0^+} E[y_i|x_i] - \lim_{x \rightarrow 0^-} E[y_i|x_i] \\ \text{(first stage)} \quad \pi_1 &= \lim_{x \rightarrow 0^+} E[D_i|x_i] - \lim_{x \rightarrow 0^-} E[D_i|x_i] \\ &= f_1(0) - f_0(0) \end{aligned}$$

Then we estimate δ_1 and π_1 and form the ratio δ_1/π_1 . Note that for a sharp RD, $\pi_1 = 1$. Otherwise, $\pi_1 = f_1(0) - f_0(0) < 1$.

Why does this work? $y_i = Y_i(1) \times D_i + Y_i(0) \times (1 - D_i)$

- From below:

$$\begin{aligned}x_i < 0 : E[y_i | x_i] &= E[Y_i(1) | x_i] f_0(x_i) + E[Y_i(0) | x_i] (1 - f_0(x_i)) \\ \Rightarrow \lim_{x \rightarrow 0^-} E[y_i | x_i] &= E[Y_i(1) | x_i = 0] f_0(0) + E[Y_i(0) | x_i = 0] (1 - f_0(0))\end{aligned}$$

- From above:

$$\begin{aligned}x_i \geq 0 : E[y_i | x_i] &= E[Y_i(1) | x_i] f_1(x_i) + E[Y_i(0) | x_i] (1 - f_1(x_i)) \\ \Rightarrow \lim_{x \rightarrow 0^+} E[y_i | x_i] &= E[Y_i(1) | x_i = 0] f_1(0) + E[Y_i(0) | x_i = 0] (1 - f_1(0))\end{aligned}$$

- Reduced form:

$$\begin{aligned}\delta_1 &= (E[Y_i(1) | x_i = 0] - E[Y_i(0) | x_i = 0]) \times (f_1(0) - f_0(0)) \\ &= E[Y_i(1) - Y_i(0) | x_i = 0] \times \pi_1\end{aligned}$$

Example in Chile revisited

- Solis (2024) estimates a fuzzy RD in which the running variable is the student's PSU score ($x_i = \text{first score} - 475$) on the **first attempt**. Whether $x_i \geq 0$, $z_i := 1[x_i \geq 0]$ serves as an instrument for $D_i = \text{ever eligible for tuition loan}$.
 - There are students who re-take the exam to get above the threshold, so it's not a sharp RD.
- His two-stage least squares regression (2SLS):

$$D_i = \pi_0 + \underbrace{\pi_1 z_i}_{\text{instrument}} + f(x_i) + \eta_i$$

$$Y_i = \beta_0 + \beta_1 D_i + h(x_i) + \nu_i.$$

The control function f in first-stage is a flexible nonlinear function: $f(x_i) = f_+(x_i) \times z_i + f_-(x_i)$. The control function in 2nd stage: $h(x_i) = h_+(x_i) \times z_i + h_-(x_i)$ with continuous h_+, h_- .

- β_1 measures the effect of having access to loans on ever enrolling in college, for students for whom the treatment status D_i does not change after the 1st PSU attempt — aka **compliers** (not always takers who will take the exams repeatedly to cross the threshold)!

Example in Chile revisited

- How to estimate f, h ? Bandwidth selection – bias-variance tradeoff, e.g. minimizing MSE via Imbens-Kalyanaram, vs. bias-corrected inference...
 - In practice, try different bandwidth and assess the robustness of your results. You will do a sensitivity analysis in PS7!
- Let's see his 2SLS results: $\hat{\beta}_1^{2sls}$ on endogenous eligibility D_i are shown.
 - Column 1: bandwidth = 44 pts, restrict to preselected students who applied for loans and were considered eligible (based on family income).
 - Column 2: bandwidth = 44 pts, restrict to nonselected students who did not apply/were ineligible (from richer families).
 - Column 3: num. observations = column 1 + column 2.
 - Column 4: bandwidth = ALL, fit f, h flexibly via a 4-th order polynomial in running variable z
- Column 5: bandwidth = 4 pts $\rightarrow$ fewer observations, estimates are quite similar!

Example in Chile revisited

TABLE 1
EVER ENROLLED IN 2 YEARS AND YEARS OF COLLEGE

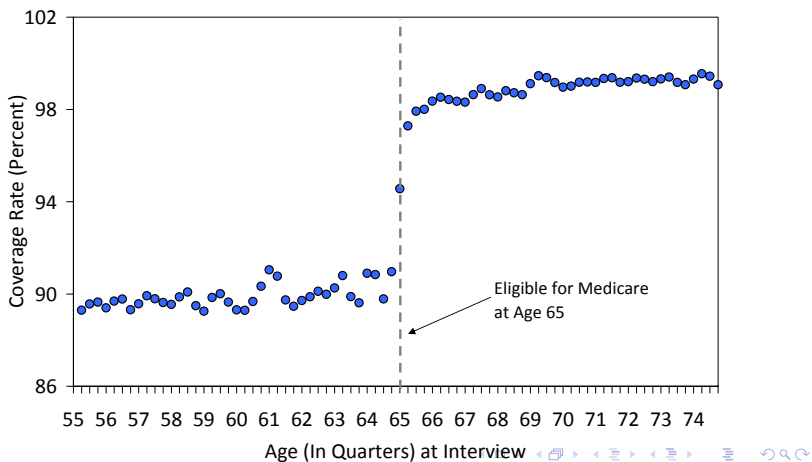
	PRESELECTED	NONSELECTED	POPULATION, COHORT 2007		
	(1)	(2)	(3)	(4)	(5)
A. Dependent Variable: Ever Enrolled in 2 Years					
Ever eligible	.158 (.016)**		.169 (.022)***	.169 (.022)***	.165 (.077)**
Preselected			.043 (.013)***	.045 (.014)***	.044 (.056)
$1(T_1 \geq \tau)$		-.006 (.012)	-.009 (.012)	-.019 (.011)*	.015 (.043)
Constant	.299 (.011)***	.275 (.008)***	.254 (.007)***	.258 (.007)***	.233 (.033)***
R^2	.112	.039	.116	.408	.059
B. Dependent Variable: Years of College					
Ever eligible	.492 (.037)***		.522 (.051)***	.509 (.052)***	.499 (.175)***
Preselected			.063 (.029)**	.074 (.032)**	.008 (.124)
$1(T_1 \geq \tau)$		-.016 (.027)	-.024 (.027)	-.040 (.026)	.024 (.097)
Constant	.769 (.026)***	.765 (.018)***	.700 (.016)***	.706 (.016)***	.690 (.075)***
R^2	.145	.043	.145	.456	.076
Observations	22,819	23,334	46,153	149,069	4,596
f specification	Linear	Linear	Linear	4th polynomial	Linear
Bandwidth	44	44	44	All	4

Example 2: Medicare

- Effect of health insurance on use of health care services.
- Most people become eligible for Medicare at age 65
- At that point, many people who previously lacked insurance get coverage
- Card, Dobkin, and Maestas study: what happens?
 - This is a **fuzzy design** because some people get on Medicare pre-65 (DI program) and some people over 65 are ineligible (or delay entering). Also, effect on insurance status is smaller than rise in Medicare enrollment.

Example 2: Medicare

Figure 1: Health Insurance Coverage Rates by Age, HIS



Example 2: Medicare

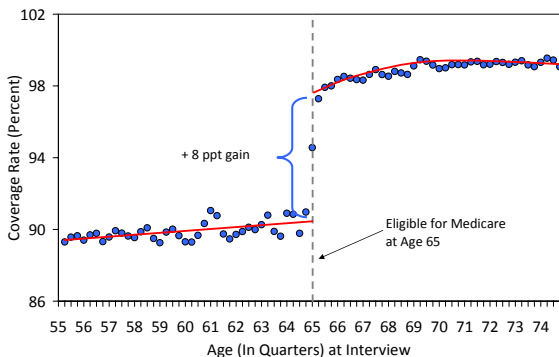
- First stage of coverage on age:
 - Running variable $x_i = \text{age} - 65$,
 - Instrument $z_i = 1[x_i \geq 0]$,
 - Endogenous treatment $D_i = \text{Medicare coverage}$.

$$D_i = \pi_0 + \underbrace{\pi_1 z_i}_{\text{jump}} + \underbrace{f_-(x_i) + z_i \times f_+(x_i)}_{\text{control for r.v.}} + \epsilon_i$$

Example 2: Medicare

- $\hat{\pi}_1 = 8$ percentage points.

Health Insurance Coverage Rates by Age, HIS

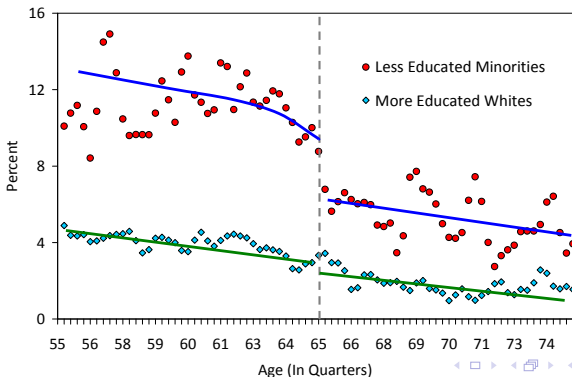


Example 2: Medicare

- Outcome: $Y_i = 1$ if did not get care for cost reasons. Reduced form regression:

$$Y_i = \pi_0 + \underbrace{\pi_1 z_i}_{\text{jump}} + \underbrace{g_-(x_i) + z_i \times g_+(x_i)}_{\text{control for r.v.}} + \epsilon_i$$

Fraction Who Did Not Get Care for Cost Reasons by Age and Group



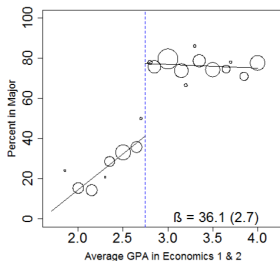
Example 3: Econ Majors

- Bleemer & Mehta “Will Studying Economics Make You Rich? A Regression Discontinuity Analysis of the Returns to College Major”:
 - UC Santa Cruz has 2.8 GPA (in econ 1-2) threshold for declaring an econ major.
 - data: 2000-2014 UCSC student records merged to National Student Clearinghouse and state unemployment insurance (UI) records.
 - research design: look for RD in prob. major in econ (first stage), and wages some years later (outcome).

Example 3: Econ Majors

- First-stage estimates of the impact of meeting the GPA threshold on economics major choice for the 2008-2012 cohorts:

Figure 1: The Effect of the UCSC Economics GPA Threshold on Majoring in Economics

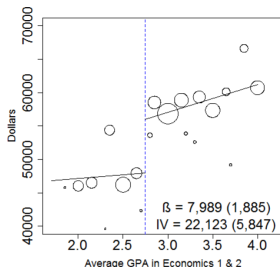


Note: Each circle represents the percent of economics majors (y axis) among 2008-2012 UCSC students who earned a given *EGPA* in Economics 1 and 2 (x axis). The size of each circle corresponds to the proportion of students who earned that *EGPA*. *EGPAs* below 1.8 are omitted, leaving 2,839 students in the sample. Fit lines and beta estimate (at the 2.8 GPA threshold) from linear regression discontinuity specification; standard error (clustered by *EGPA*) in parentheses.

Example 3: Econ Majors

- Reduced-form estimates of the impact of meeting the GPA threshold on wages:

Figure 2: The Effect of the UCSC Economics GPA Threshold on Annual Wages



Note: Each circle represents the mean 2017-2018 wages (y axis) among 2008-2012 UCSC students who earned a given *EGPA* in Economics 1 and 2 (x axis). The size of each circle corresponds to the proportion of students who earned that *EGPA*. 2017-2018 wages are the mean EDD-covered California wages in those years, omitting zeroes. Wages are CPI-adjusted to 2018 and winsorized at 2% above and below. *EGPAs* below 1.8 are omitted, leaving 2,446 students with observed wages. Fit lines and beta estimate (at the 2.8 GPA threshold) from linear regression discontinuity specification and instrumental variable specification (with majoring in economics as the endogenous variable); standard errors (clustered by *EGPA*) in parentheses. Sources: The UC-CHP Student